Track 1 Smart Campus · Linear energy model with test-aware feature imputation

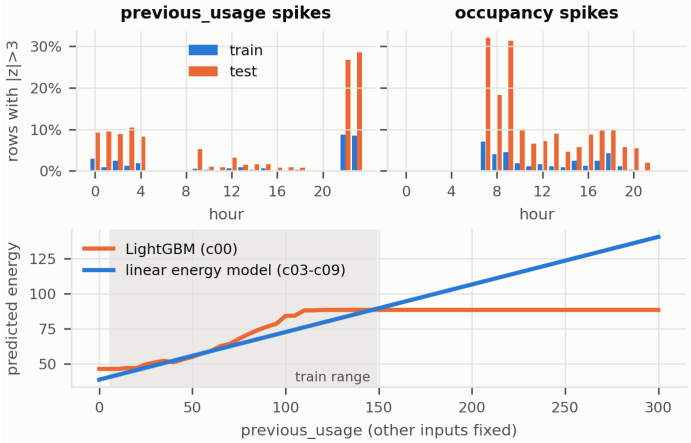
Team Cookie Squad · Final model c09 · Public RMSE **3.20392** (rank 1; c04 3.33681, c03 3.36875) · energy-model CV RMSE 3.048 clean / 3.228 test-like

1 · Problem

Predict *energy_usage* for 12 buildings (8 types) from building, hour / weekday / month, temperature, humidity, occupancy and *previous_usage*; metric RMSE. Test rows arrive shuffled without ids, so features are row-local. *previous_usage* is not the previous hour: its daily profile matches energy at the **same** hour (residential corr. 0.97 vs 0.52 one hour back) and rows never chain, so it is a same-slot reference from an earlier period.

2 · Critical insight: test is shifted, and the extremes are real

Test has 3× train's missing rate (18.2% vs 5.8% of rows) and, by a target-free consistency model (each feature predicted from the others), **16.6%** spike rows (|z|>3) vs **4.1%** in train, on a schedule (prev 22:00–04:00, occupancy 07:00–09:00). They are not errors: on train spike rows the energy left unexplained without the feature tracks its deviation (r = 0.92) along the ordinary-row line. So the energy equation must **extrapolate**; trees saturate.



3 · Validation

Energy model: paired 5-fold × 4-repeat CV scored **clean**, **test-like** (validation rows get missing-value masks sampled from the test file's joint patterns) and **shift-weighted** (spike rows ×4.05 to test frequency; a train-vs-test classifier, AUC 0.70, independently gave them ~3×). Imputers: a **label-free proxy** hides observed values in held-out public-test rows, re-imputes them and weights each error by that feature's energy slope, i.e. the energy error caused by imputation, using no targets.

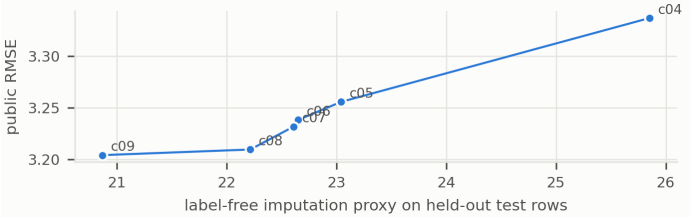
4 · Method

Energy model (ridge, α=3): per-building intercept, occupancy and temperature slopes; shared *previous_usage* (~0.34) and humidity slopes; smooth daily profile per type with shrunk hour corrections; weekend; season; temperature². **Missing inputs:** c03's per-pattern sub-models distort correlated slopes (without occupancy the prev slope inflates 0.34→0.46); **c04** imputes inputs with LightGBM and keeps the **full** model. **c05–c09** make the imputers test-aware: a train/test domain flag (predict as test), an average of 3 settings, test rows ×2, 5 copies of test-like mask augmentation, and **deviation features** (each observed input minus its building×hour and building mean), because spikes co-occur across features. LightGBM only fills inputs; the energy prediction stays linear.

5 · Results

RMSE	Clean CV	Test-like	Shift-wtd	Public
c00 LightGBM	3.341	3.964	4.281	–
c03 linear	3.077	3.296	3.543	3.36875
c04 + imputation	3.048	3.228	3.363	3.33681
c05 domain flag				3.25549
c08 + mask aug ×5				3.20937
c09 final				**3.20392**

c04 beats c03 in 4/4 CV repeats on every view (test-like Δ -0.068, shift-weighted -0.180; MSE 95% CI [-0.67, -0.25]). Test-aware imputation then cut public RMSE 3.33681 → 3.20392. The label-free proxy ranked all six submissions in public-score order (below); its magnitude is less reliable, so final steps within public noise were not pursued.



6 · Error analysis: what is left

Rows without missing values (82% of test) are at the noise floor (~3.0): LightGBM, ExtraTrees, CatBoost and an MLP trained on out-of-fold residuals reach R² below 0, and group slopes, spike-specific slopes, density weighting and blends did not help. Forensics found no sequence, duplicates or cross-row linkage. The remaining error sits in rows with a missing value, where a spike hidden in the missing input cannot be recovered. c09 also beats c04 on train-like held-out rows (proxy 12.2 vs 13.0).

7 · Campus insights (associations, not causal effects)

+100 occupants ~ **+13.4** energy units in the lecture hall vs **+4.6** in residences. Science buildings carry the largest modelled base load (39–43 vs 16–27), consistent with fixed equipment loads (interpretation). Residences step down at 08:00 and up at 21:00. Uses: hourly peak planning and flagging buildings that exceed expected use.

8 · Limitations

Spikes beyond the labelled range are extrapolated linearly (supported by train spikes, assumed beyond). c05–c09 assume the private test resembles the public test; if not, c09 is still better than c04 on train-like rows. No dates, so no external weather or calendar data. Inference: numpy / pandas / LightGBM 4.5.0 from the base image, about 3 s.